



Mahindra University Hyderabad
École Centrale School of Engineering
Minor-II exam

Program: B. Tech.

Branch: CM

Year: II

Semester: II

Subject: Foundational Mathematics for ML (MA2221)

Date: 15/04/2026

Start Time: 2:00 PM

Time Duration: 1.5 Hours

Max. Marks: 20

Instructions:

- 1) All questions are self-explanatory; no clarification will be provided during the exam.
- 2) Use of one non-programmable scientific calculator is allowed.

1. Answer any two of the following.

- (a) (3 points) Describe the problem of overfitting in model learning.
- (b) (3 points) Explain K -fold cross-validation.
- (c) (3 points) Explain how real-world data can be represented as vectors.

2. (4 points) Consider the following multivariate mathematical function:

$$f(x, y) = (x^2 + y) \cdot (2x - y^2)$$

Perform the reverse sweep of automatic differentiation to compute the exact gradients $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(1, 2)$.

3. (4 points) For $\min_x \frac{1}{2}x^T Qx + c^T x$ subject to $Ax \leq b$ with $Q \succ 0$, form the Lagrangian and derive the dual problem.
4. (2 points) Compute the convex conjugate $f^*(s)$ for $f(x) = \frac{1}{2}x^2$ on $\mathbb{R}$.
5. (4 points) Let $\mathbf{X} = [X_1, X_2]^T$ be a bivariate Gaussian with:

$$\mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

- (a) Find the marginal distribution of X_1 .
- (b) Find the conditional distribution $p(X_1|X_2 = 3)$.
- (c) Are X_1 and X_2 independent?